MIL-B-50999 (MU)
7 December 1971

MILITARY SPECIFICATION

BOOSTER, AUXILIARY, M122, METAL PARTS
FOR, AND ASSEMBLY AND LOADING OF

1. SCOPE

1.1 This specification covers the metal parts, assembly and loading for the M122 Auxiliary Booster used in the M567 Fuze.

2. APPLICABLE DOCUMENTS

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card, Military Standard
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

COMMERCIAL

ASTM-STD-31 - Physical and Mechanical Testing of Metals, Non-Destructive

FSC: 1375

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ORDNANCE DRAWINGS

7548040 - Carton, Packing, Ammunition for Booster, Auxiliary, M122 Assembly; Detents, Packing and Marking
7548041 - Box, Packing, Ammunition for Booster, Auxiliary, M122 Assembly, Details and Packing
9239270 - M122 Auxiliary Booster Assembly

PUBLICATIONS

ET-9239270 - Equipment Tabulation

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts and Assemblies.-The parts and assemblies shall comply with all requirements specified on Drawing 9239270 and with all requirements specified in applicable specifications.

3.3 Density.-The density after consolidation shall be between 1.55 and 1.72.

3.4 Cracks or Splits (dwg. 9239271).-No cracks or splits shall be permitted in the metal container walls when examined as specified in 4.3.3.

3.5 Moisture Content of RDX.-The unconsolidated RDX shall have a moisture content no greater than 0.50 percent immediately prior to loading.

3.6 Workmanship.-All parts and assemblies shall be fabricated in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust, and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall any parts be contaminated by the cleaning agent. Exterior surface coatings shall be uniform and continuous except for a few light scratches not exposing base material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements

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as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Production.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with indentification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and the certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions.

4.2.1 Lot Formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quanitity of units from which the sample was drawn and shall not

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be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

b. RDX from one batch or blending of batches which results in homogeneity of all explosives used in a loaded lot.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accccordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Cup, Auxiliary Booster (see dwg. 9239271 covering a detail of dwg. 9239270)

Categories	Defects	Method of Inspection
Critical:	None defined.	

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Major:	AQL 1.0 percent	
101.	Distance to flange	Gage
102.	Outside diameter	Gage
103.	Outside diameter of flange ..	Gage
104.	Concentricity of hemisphere .	Gage
105.	Thickness of flange	Gage
106.	Depth of hemisphere	Gage
Minor:	AQL 1.0 percent	
201.	Presence of coating	Visual
202.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.2 M122 Auxiliary Booster Assembly (see dwg. 9239270).

Categories	Defects	Method of Inspection
Critical:		
1.	Presence of RDX on outside of booster cup	Visual - (5X Mag. with dazor light)
Major:	AQL 0.65 percent	
101.	RDX above flush	Gage
102.	Outside diameter, max.	Gage
103.	Depth of hemisphere, min. ...	Gage
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.3 Packing Carton (see F7548040).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Presence of seal	Visual
102.	Marking illegible or missing.	Visual
Minor:	None defined.	

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4.2.2.4 Packing Box (see F7548041).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
	101. Damaged box	Visual
	102. Marking missing or illegible .	Visual
Minor:	None defined.	

4.2.3 Testing

4.2.3.1 Density - Auxiliary Booster Assembly (dwg. 9239270) - Major Defect - Improper Density. - The sample plan for this test shall be in accordance with Code Letter N and AQL of 0.65 percent.

4.2.3.2 Cup Integrity

4.2.3.2.1 Cup Integrity Before Loading Cup, Auxiliary Booster (dwg. 9239271) - Major Defect.-The sample plan for this test shall be in accordance with Code Letter N and AQL of 0.65 percent.

4.2.3.2.2 Cup Integrity After Loading Booster, Auxiliary, M122 (dwg. 9239271 covering a detail of dwg. 9239270) - Major Defect.-The sample plan for this test shall be in accordance with Code Letter N and AQL of 0.65 percent.

4.2.4 Inspection Equipment. - Equipment Tabulation Number ET-9239270 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. Those equipment lists coded F on the ET require the contractor to supply designs for inspection equipment per 6.2.

4.2.4.1 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.2.4.2 Government Use of Contractor's Inspection Equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.2.4.3 Supply and Maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with MIL-I-45607.

4.3 Test Methods and Procedures

4.3.1 Density.-The test shall be performed on the M122 Auxiliary Booster (dwg. 9239270). The assembly and an unfilled cup (dwg. 9239271) shall both be weighed on a balance and the weight recorded as dry weight. The assembly and an unfilled cup shall then be weighed while completely submerged in water. The temperature of the water shall be 68° Fahrenheit (F) plus or minus 2°F. This weight shall be recorded as submerged weight. (When submerged, the position of the sample shall be such that no air is entrapped.) Density shall be calculated as follows:

$$\text{Density} = \frac{A-C}{(A-C) - (B-D)}$$

where:

A = Dry weight of assembly
B = Submerged weight of assembly
C = Dry weight of empty cup
D = Submerged weight of empty cup

The result shall be converted to 4°C using appropriate tables. Equipment shall be as specified in 4.2.4.

4.3.2 Cup Integrity

4.3.2.1 Cup Integrity Before Loading.-This test shall be performed on the cup, auxiliary booster (dwg. 9239271) before loading. The cups shall be subjected to a fluorescent liquid penetrant inspection as specified in ASTM-STD-3i, E165-65 and examined internally and externally under 5X magnification. Any sample which fails to meet the requirements of 3.4 shall be classified defective. Equipment shall be as specified in 4.2.4.

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4.3.2.2 Cup Integrity After Loading.-This test shall be performed on the M122 auxiliary booster (dwg. 9239270). Test procedure and requirements are as specified in 4.3.2.1 with the exception that inspection is conducted on exterior surfaces only. Equipment shall be as specified in 4.2.4.

4.3.3 Moisture Content of RDX.-The sampling and testing procedures for determining the moisture content of the RDX shall be in accordance with MIL-STD-398.

5. PREPARATION FOR DELIVERY

5.1 Packing.-The loaded auxiliary cup shall be packed and marked in accordance with dwg's 7548040 and 7548041.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of Paragraphs 6.2.1 through 6.2.5.

6.2.1 Inspection Equipment Lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock Numbers of existing equipment designs, or codes as indicated in Paragraph 6.2.2. Design action required of the contractor is described in Paragraphs 6.2.3 and 6.2.4. The contractor will be required to prepare detailed drawings in accordance with 6.2.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.2.5.

6.2.2 Inspection Equipment List Codes.-The inspection equipment, as defined in 6.2.3 and 6.2.4, will be designated in the EL by the following codes:

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CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design; mandatory for use.

6.2.3 Army Designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.2.5.

6.2.4 Contractor Designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. For inspection equipment designs, the contractor shall refer to MIL-HDBK-204. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.2.5.

6.2.5 Submission of Contractor Designs.-All submitted designs shall contain a reference to the paragraph number and defect number in Section 4 of this specification and the appropriate component or

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assembly drawing number and revision letter to which the specific design applies. Unless otherwose specified on the EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-F. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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